

Network Incident Analysis: TCP SYN Flood

Simulated Case Study

Incident Summary

The web server experienced a denial-of-service (DoS) attack consistent with a TCP SYN flood. The packet capture shows a large number of SYN requests from 203.0.113.0 being sent to the server at 192.0.2.1.

The server responds with SYN-ACK packets, but many of the connections are never completed. As these unfinished connections build up, server resources are consumed and legitimate users begin experiencing slow responses and connection timeouts.

Technical Analysis and Impact

A normal TCP connection uses a three-way handshake: the client sends SYN, the server replies with SYN-ACK, and the client sends ACK to complete the connection.

During this incident, the server repeatedly receives SYN requests and sends SYN-ACK responses, but the expected final ACK does not arrive for many of the connections. Those connections remain half-open and continue using resources while the server waits for them to finish.

With enough half-open connections, fewer resources remain available for legitimate users. The packet evidence and resulting timeout behavior are consistent with a SYN flood, which can eventually make the company's website or sales page unavailable if the traffic continues.